Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 06



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Recap of Previous Lecture



Topic Reflexive relation and irreflexive relation

2^{n^2-n} $a^R a, \forall a \in A$ 2^{n^2-n} $a \not R a, \forall a \in A$

Topic Symmetric, anti-symmetric and asymmetric relation

if $a^R b$ then $b^R a$ $\forall a,b \in A$

if $(a^R b \& b^R a)$ then $a = b$ $\forall a,b \in A$

if $a^R b$ then $b \not R a$ $\forall a,b \in A$

Slide

Topics to be Covered



- Topic: Symmetric, anti-symmetric and asymmetric relation
- Topic: Transitive relation
- Topic: Complement of a relation
- Topic: Inverse of a relation
- Topic: Composite of two relations

Slide



Topic : Types of Relations

All this relations are defined from set A to same set A {i.e. on set A}

1. Diagonal Relation (Identity relation)
2. Reflexive Relation
3. Irreflexive Relation
4. Symmetric Relation
5. Anti-symmetric Rel^{n}
6. Asymmetric Rel^{n}
7. Transitive Relation

8. Complement of a relation
9. Inverse of a relation
10. Composite relation

Slide



Topic : Diagonal Relation (Identity Relation)

Diagonal relation on set A is denoted by Δ_A. and it is defined as,

$$\Delta_A = \{(a,a) \mid a \in A\}$$

it is definition of a set

eg: let $A = \{1,2,3\}$

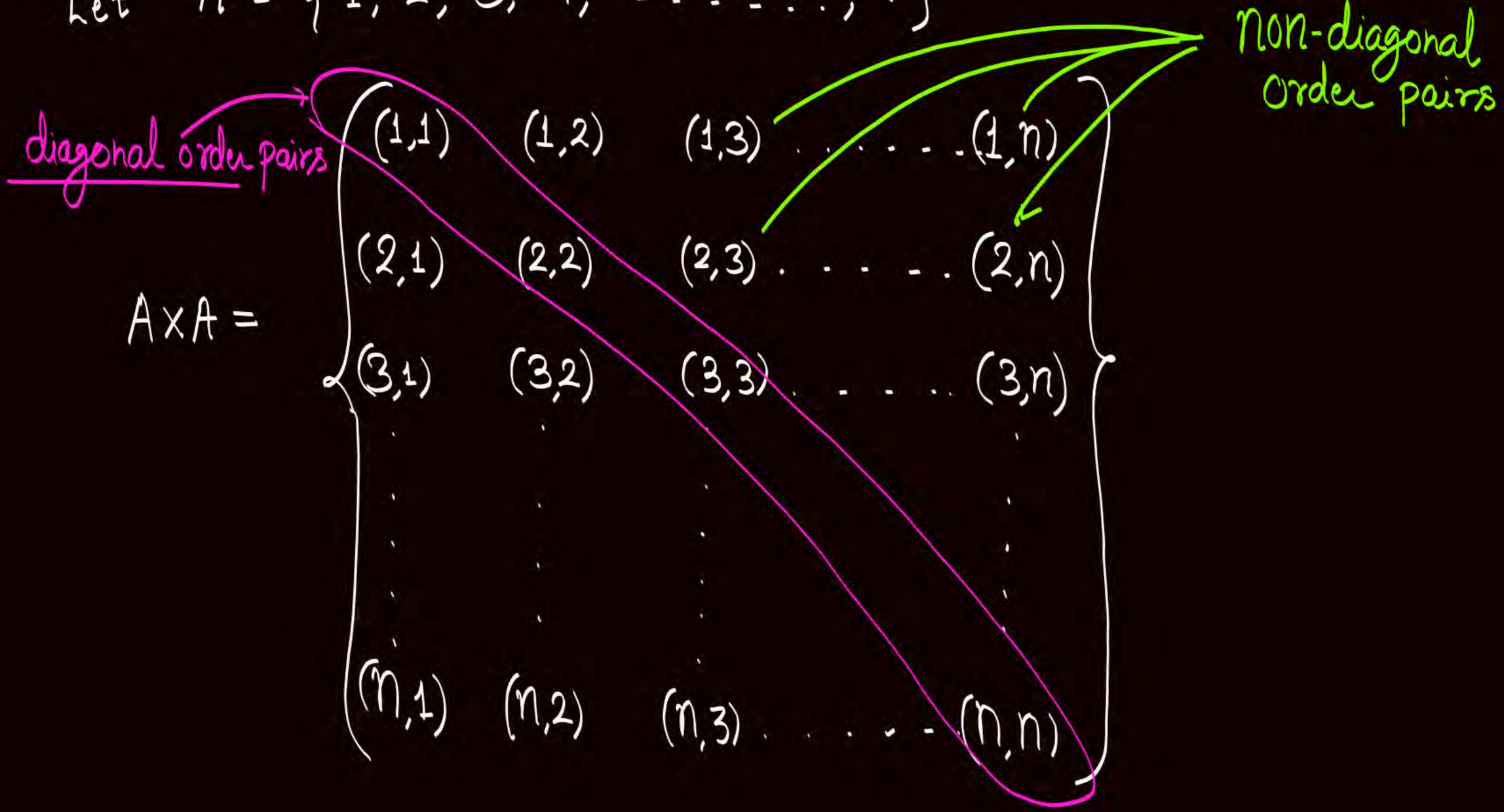
$R_1 : \{(1,1),(2,2)\}$ $(3,3) \notin R_1$ $\therefore R_1$ is not a diagonal relation on set A.

$R_2 = \{(1,1),(2,2),(3,3),(1,2)\}$ + order pair (1,2) can never be an element of diagonal rel^n $\therefore R_2$ is not a diagonal rel^n.

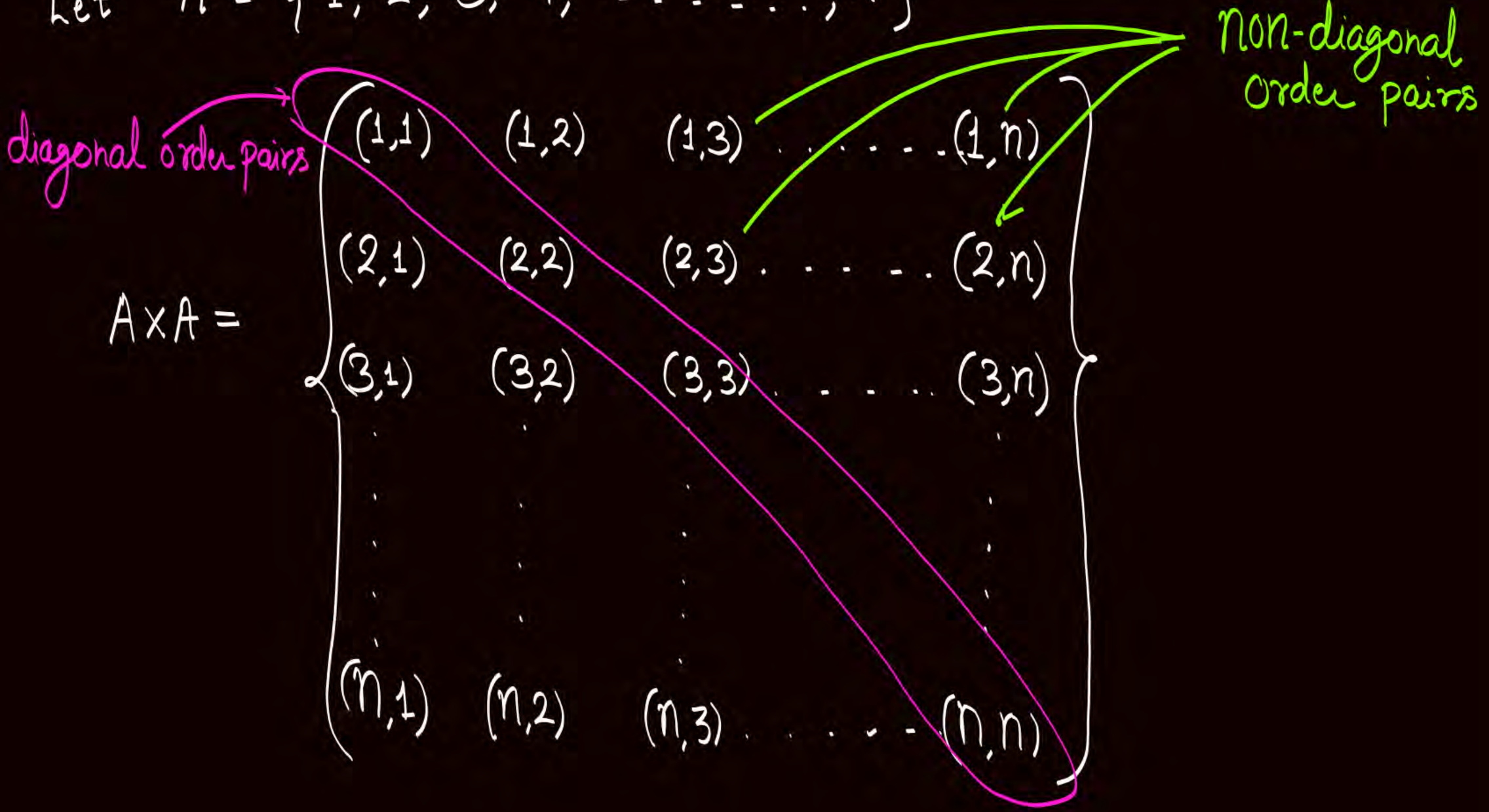
$R_3 = \Delta_A = \{(1,1),(2,2),(3,3)\}$

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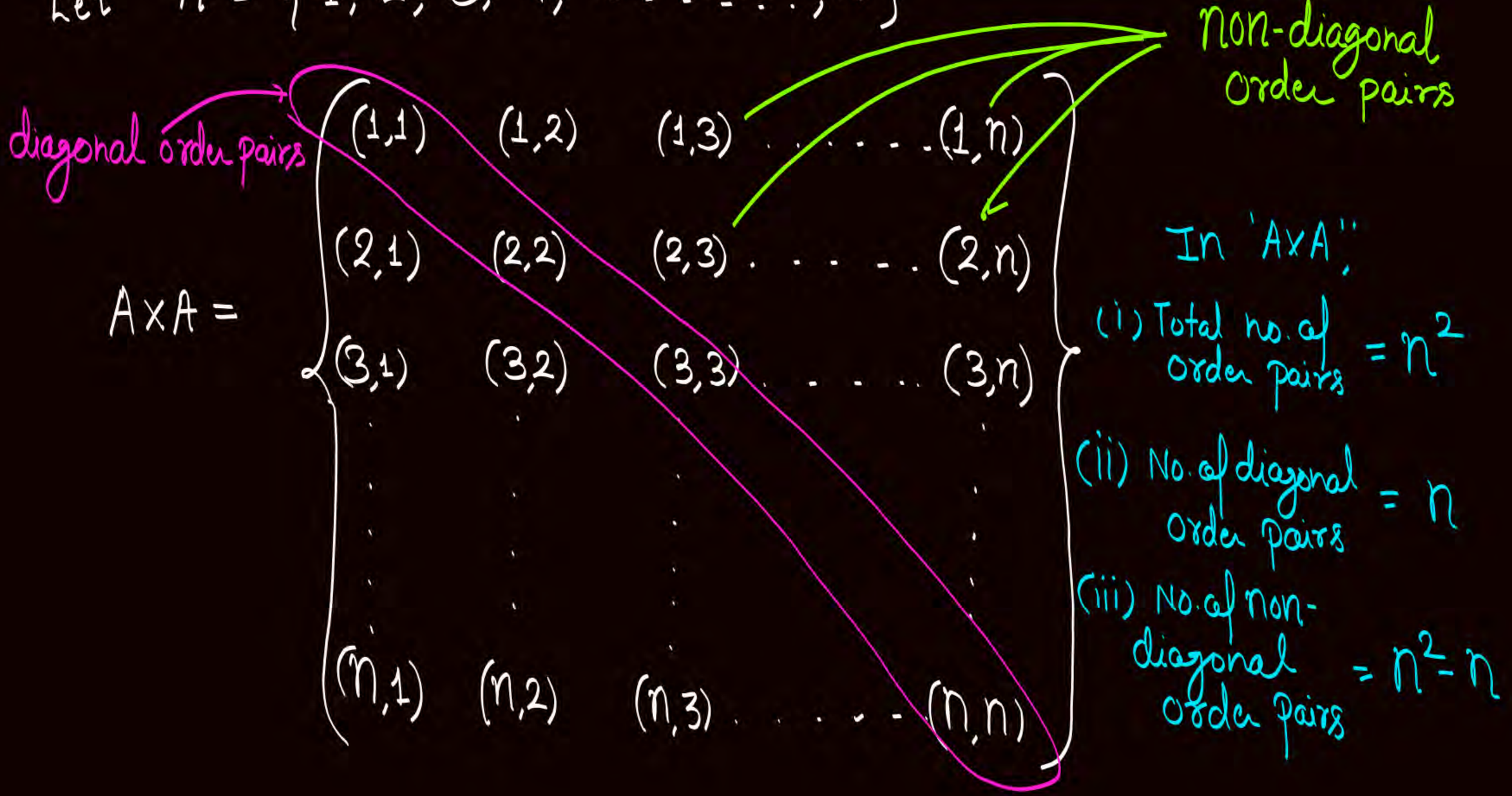
Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$



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In 'A x A':

(i) Total no. of order pairs $= n^2$

(ii) No. of diagonal order pairs $= n$

(iii) No. of non-diagonal order pairs $= n^2 - n$



Topic : Reflexive Relation

{ All diagonal order pairs must be present }

A relation R on set A is called reflexive relation if and only if,

$$a \, R \, a, \quad \forall a \in A$$

i.e. $(a,a) \in R, \quad \forall a \in A$

it is a Constraint for a relation to be reflexive

$a \, R \, a \Rightarrow$ a relates to a w.r.t. relation R

i.e. $(a,a) \in R$

$a \not R \, a \Rightarrow$ 'a' does not relate with 'a' w.r.t. relation R

i.e. $(a,a) \notin R$

Slide

eg: Let $A = \{1, 2, 3\}$

$R_1 = \{(1,1), (2,2), (3,3)\}$ { R_1 is a diagonal relation as well as reflexive relation }

$R_2 = \{(1,1), (2,2), (3,3), (2,3)\}$ { R_1 is a reflexive relation, but it is not a diagonal relation }

$R_3 = \{(1,1), (2,2)\}$ $(3,3) \notin R_3$ ∴ Not a reflexive relation on set A.

$R_4 = \{(1,1), (2,2), (1,3)\}$ Not a reflexive relation on set A.

Note: ① Every diagonal relation on set A is also a reflexive relation on set A, but every reflexive rel^{n} on set A need not be a diagonal rel^{n} on set A.

② Diagonal relation on set A is the smallest reflexive relation on set A.

③ Largest reflexive relation on set A is "A×A"

eg: Relation "$\leq$" is a reflexive relation on any set of real numbers.

Let $A = \{1, 2.5, 3\}$

$(a, b) \in \text{"}\leq\text{"}$ iff $a \leq b$

$\leq = \{(1,1), (1,2.5), (1,3)$

$(2.5,1), (2.5,2.5), (2.5,3)$

$(3,1), (3,2.5), (3,3)\}$

All diagonal order pair are Present, ∴ Reflexive Relⁿ.

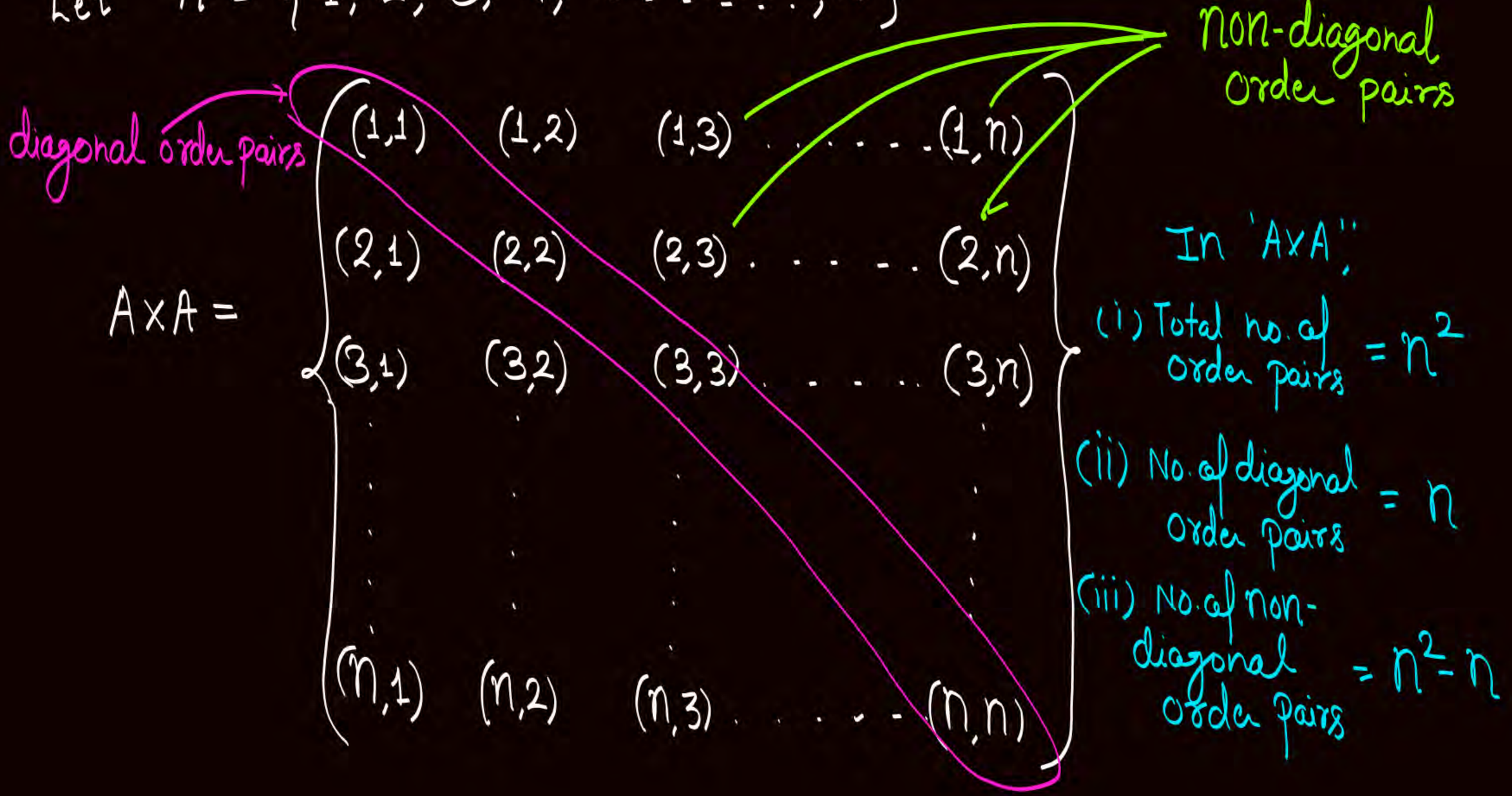
eg: Relation "$\div$" (divides) is a reflexive relation on any set of non-zero positive integers. { $(a,b) \in$ "$\div$" iff a divides b }

eg: Relation "$\subseteq$" (subset) is a reflexive relation on any Collection of sets. { Every set is a subset of itself,
∴ Every set will relate with itself
hence reflexive }

Q: Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

How many reflexive relation are possible on set A.

Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$



In 'A x A':

(i) Total no. of order pairs $= n^2$

(ii) No. of diagonal order pairs $= n$

(iii) No. of non-diagonal order pairs $= n^2 - n$

Q: Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

How many reflexive relation are possible on set A. {0 to all}

Reflexive Relation = (All diagonal order Pairs must be present) and (apart from that any number of non-diagonal order pairs may be Present)

∴ Number of Reflexive Relation = $^{n}C_{n} * \{ {}^{n^2-n}C_0 + {}^{n^2-n}C_1 + {}^{n^2-n}C_2 + \cdots\cdots + {}^{n^2-n}C_{n^2-n} \}$

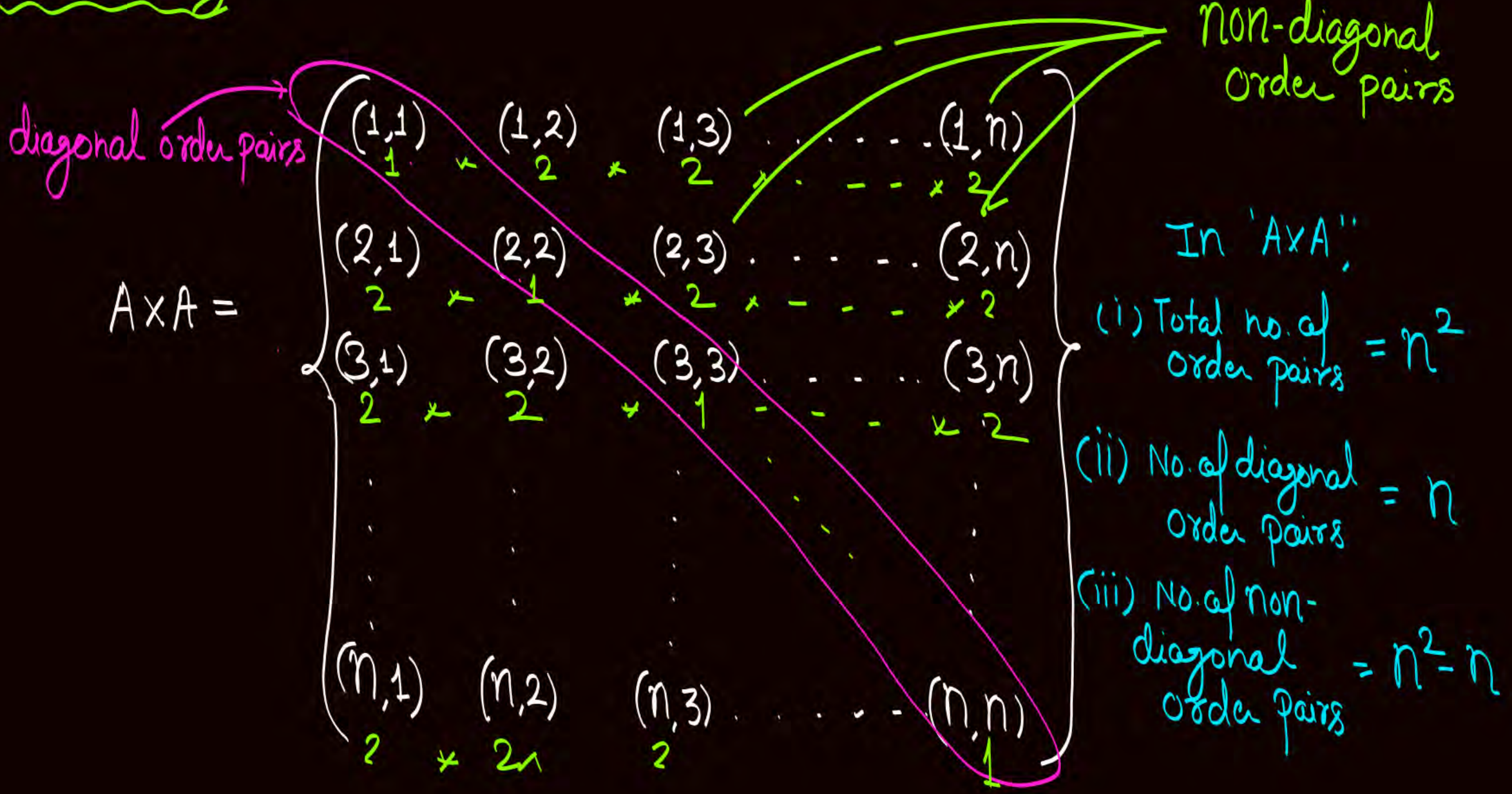
Choose all 'n' diagonal order pairs

Choose any no. of non-diagonal order pairs out of (n^2-n)

$= 1 * 2^{n^2-n}$

$= 2^{n^2-n}$

Another Way :-



In 'A × A' :

(i) Total no. of order pairs $= n^2$

(ii) No. of diagonal order pairs $= n$

(iii) No. of non-diagonal order pairs $= n^2 - n$



Topic : Irreflexive Relation

{ none of the diagonal order pair should be present }

A relation R on set A is called irreflexive relation if and only if.

$$a \not R a, \quad \forall a \in A$$

i.e $(a,a) \notin R, \quad \forall a \in A$

Slide

eg: Let $A = \{1, 2, 3\}$

$R_1 = \{\ \}$ Empty relation

Empty relation on any set A is the smallest irreflexive relation on set A.
{it is not reflexive}

$R_2 = \{(1,2), (3,1), (2,3)\}$ { No order of type (x,x) is irreflexive relation }
{it is not reflexive}

$R_3 = \{(1,2), (2,3), (3,3)\}$

$(3,3) \in R_3$ s.t. $3=3$ ∴ Not an irreflexive relation
{neither reflexive nor irreflexive}

Note:

① Empty relation is the smallest irreflexive relation

② let $|A| = n$, then largest irreflexive relation on set A will contain $(n^2 - n)$ order pairs.

③ There may be some relations on set A which are neither reflexive nor irreflexive

Q: Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

How many irreflexive relation are possible on set A.

Q: Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

How many irreflexive relation are possible on set A. {0 to all}

Irreflexive Relation = (None of the diagonal order Pairs should be present) and (apart from that any number of non-diagonal order pairs may be Present)

Choose Zero diagonal order pairs

Choose any no. of non-diagonal order pairs out of (n^2-n)

$\therefore$ Number of irreflexive Relation $= {}^{n}C_0 * \{{}^{n^2-n}C_0 + {}^{n^2-n}C_1 + {}^{n^2-n}C_2 + \cdots\cdots + {}^{n^2-n}C_{n^2-n}\}$

$= 1 * 2^{n^2-n}$

$= 2^{n^2-n}$



Topic : Symmetric Relation

A relation R on set A is called symmetric relation if and only if,

(if $a \, R \, b$ then $b \, R \, a$, $\forall a,b \in A$)

{ $\forall a,b \in A$, if - then statement should be true }

i.e., if $(a,b) \in R$ then $(b,a) \in R$, $\forall a,b \in A$

Slide

eg: let $A = \{1, 2, 3\}$

- $R_1 = \{\ \}$ it is smallest symmetric relation on set A.

Empty relation

$R_2 = \{(1,1)\}$ { w.r.t. diagonal order pair $a = b$
∴ if (a,b) is true then (b,a) is automatically true
Hence because of Presence or absence of diagonal order pair there will be no Problem w.r.t symmetric property }

Hence R_2 is a Symmetric relation.

$R_2 = \{(1,1), (1,2), (2,1)\}$

diagonal order pair ∴ No problem

→ both $(1,2) \& (2,1) \in R_3$ ∴ No problem

R_3 is Symmetric Relation

$R_4 = \{ (1,1), (2,3), (3,1), (3,2) \}$

$(3,1) \in R_4$ but $(1,3) \notin R_4$

if is true then is false

Cond^n is false for at least one order pair

$\therefore R_4$ is not symmetric Relⁿ.

$R_5 = A \times A$ it is the largest symmetric relation on set A.



Topic : Anti-Symmetric Relation

* A relation R on set A is called anti symmetric relation if and only if,

if $(a \, R \, b$ and $b \, R \, a)$ then $a = b$ $\forall a, b \in A$

i.e., if $((a,b) \in R$ and $(b,a) \in R)$ then $a = b$, $\forall a, b \in A$

"$a = b$" means diagonal order pair

"if" if Cond^n is true then order pair must be diagonal order pair,

otherwise, if Cond^n itself must be false

Slide

Note: In anti-symmetric relation there will be no problem because of presence or absence of diagonal order pairs because w.r.t. diagonal order pair "a=b" will always be true ∴ if-then statement will be true w.r.t. diagonal order pair.

Note: In case "a≠b" we know that then condⁿ is false, ∴ if "then" condⁿ is false then for statement to be true "if" statement should be false

eg. let $A = \{1, 2, 3\}$

$R_1 = \{\ \}$ it is the smallest anti-symmetric relation

Empty rel^n

$R_2 = \{(1,1)\}$ (a = 1, b = 1)

$(a,b) \in R_2$ & $(b,a) \in R_2$

but $1 = 1$ i.e. $a = b$

∴ No problem

Presence or absence of diagonal order pair does not create any problem w.r.t. anti-symmetric Rel^n.

$R_3 = \{(1,1), (1,3), (2,3)\}$

$(1,3) \in R_3$ but $(3,1) \notin R_3$ } ∴ if cond^n itself is false we don't need to worry about them

$(2,3) \in R_3$ but $(3,2) \notin R_3$ } ∴ it itself is false

Rel^n R_3 is anti-symmetric

R_4: $\{(1,2), (3,1), (2,3), (3,2)\}$

$(2,1) \notin R_4$ $(1,3) \notin R_4$

$(2,3) \in R_4$ & $(3,2) \in R_4$ ← if is true

but $2 \neq 3$ ← then is false

∴ if-then false

Hence, Not anti-symmetric



Topic : Asymmetric Relation

A relation R on set A is asymmetric relation if and only if,

if $a \, R \, b$ then $b \not R a$, $\forall a, b \in A$

i.e; if $(a,b) \in R$ then $(b,a) \notin R$, $\forall a, b \in A$

it simply says if $(a,b) \in R$ then (b,a) must not belong to R no matter $a=b$ or not. i.e; even diagonal order pairs are not allowed in asymmetric relation

Slide

eg: let $A = \{1, 2, 3\}$

$R_1 = \{\ \}$ ← Smallest asymmetric rel^n on set A.

Empty relation

$R_2 = \{(1,1)\}$ — diagonal order pairs are not allowed in asymmetric relation

$R_3 = \{(1,2), (3,1), (1,3)\}$

$(2,1) \notin R_3$ ∴ ok

$(3,1) \in R_3$ as well as $(1,3) \in R_3$

∴ Not asymmetric

$R_4 = \{(1,2), (3,1), (3,3)\}$ Not asymmetric

(3,3) ✗ Not allowed

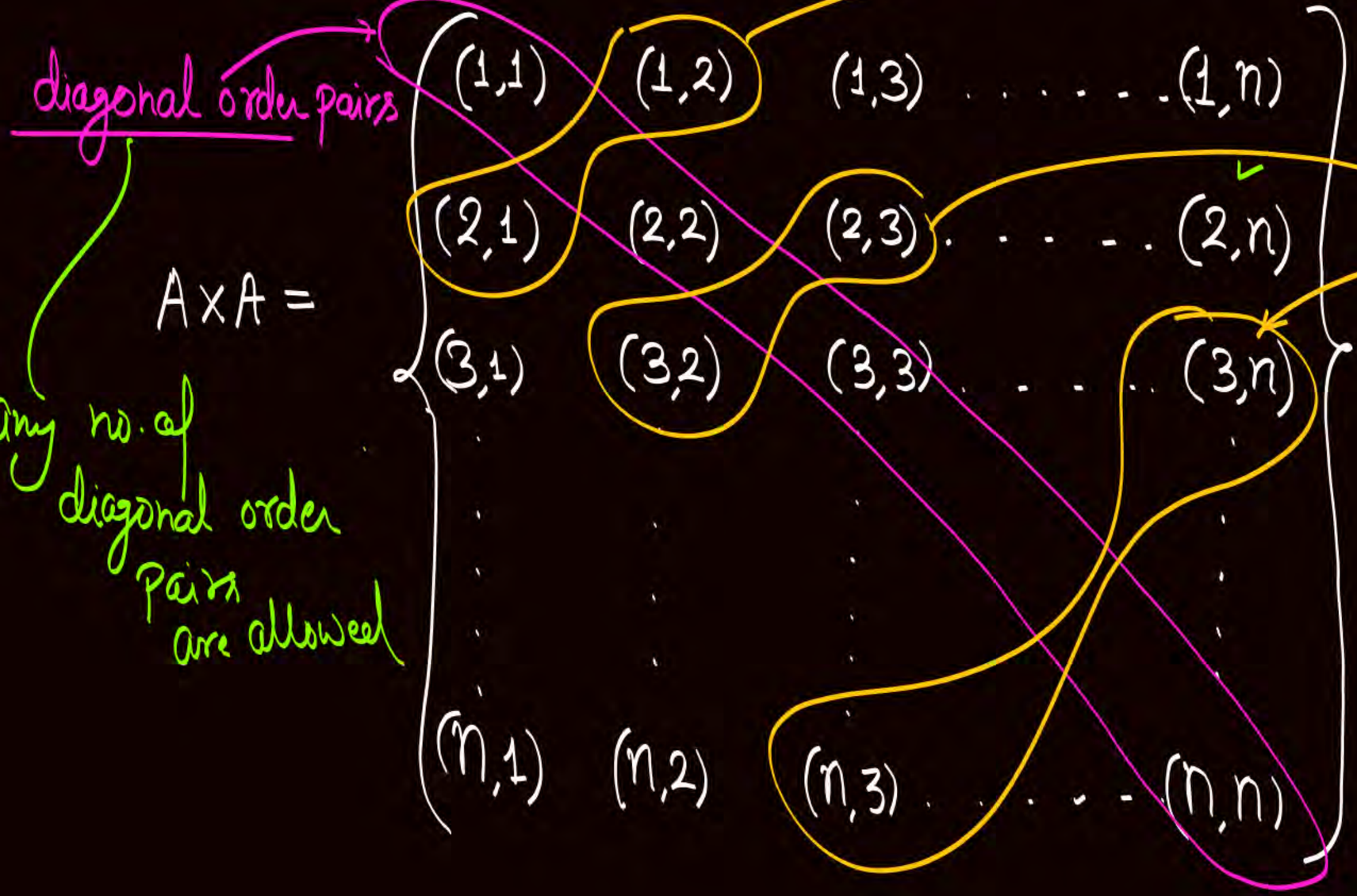
$R_5 = \{(1,2), (1,3), (2,3)\}$ it is an asymmetric relation.

H.W.

Q: Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

How many Symmetric relation are possible on set A.

Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$



Number of Symmetric Pairs of non-diagonal order Pairs $= \left(\frac{n^2-n}{2}\right)$

Pairs of Symmetric Non-diagonal order Pairs.

↓

Either both order pairs of Symmetric Non-diagonal pair will be present or None of them

Symmetric Relation = Any number of diagonal order pairs and Any number of Symmetric pairs of non-diagonal order pairs.

Number of Symmetric Rel^n $= 2^n * 2^{\left(\frac{n^2-n}{2}\right)}$

$= 2^{\left(\frac{n^2+n}{2}\right)}$

Ans

Another approach:→

- In a non-diagonal order pair (a, b) we know $a \neq b$ i.e a & b are distinct.
- For two distinct elements 'a' & 'b' possible combination of order pairs of a & b that may be present in a relation are,

(i)	(ii)	(iii)	(iv)
both (a,b) & (b,a) are present	(a,b) is present but (b,a) is absent	(a,b) is not present but (b,a) is present	Neither (a,b) is present nor (b,a) is present
(a,b), (b,a)	(a,b) ~~(b,a)~~	~~(a,b)~~, (b,a)	~~(a,b)~~, ~~(b,a)~~

Four possible choices w.r.t. order pairs of two distinct element

Another approach:→

- In a non-diagonal order pair (a, b) we know $a \neq b$ i.e a & b are distinct.
- For two distinct elements 'a' & 'b' possible combination of order pairs of a & b that may be present in a relation are,

(i) both (a,b) & (b,a) are present

(a,b), (b,a)

Allowed w.r.t Symmetric Relⁿ

(ii) (a,b) is present but (b,a) is absent

(a,b) ✓ (b,a) ✗

(iii) (a,b) is not present but (b,a) is present

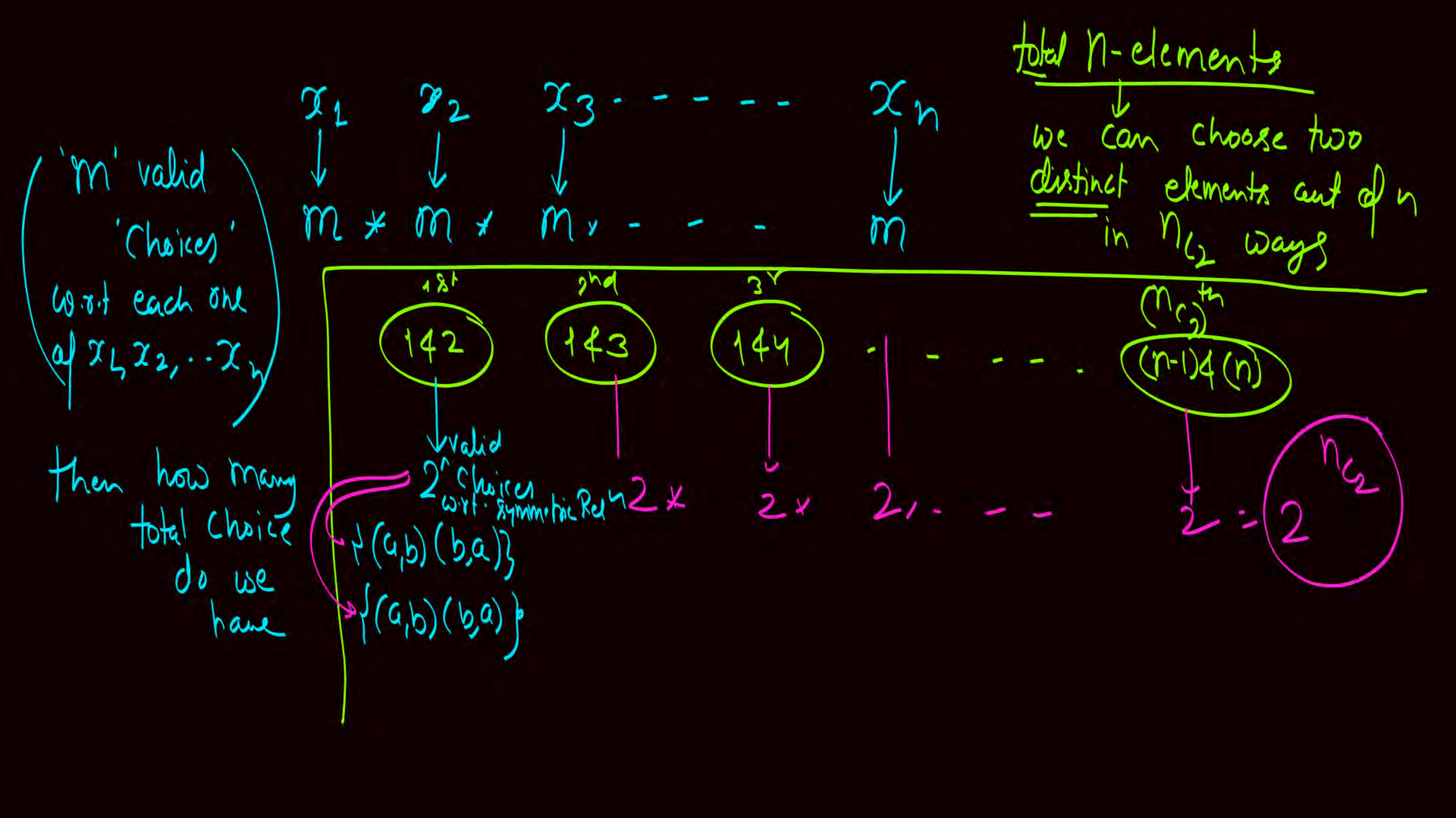
(a,b) ✗ , (b,a) ✓

This two combinations of order pairs are not allowed in a Symmetric Relⁿ

(iv) Neither (a,b) is Present nor (b,a) is present

(a,b) ✗ , (b,a) ✗

Allowed w.r.t Symmetric Relⁿ


total n-elements
we can choose two distinct elements out of n in n_{C_2} ways
x_1 x_2 x_3 ------ x_n
m * m * m * - - - m
('m' valid 'choices' w.r.t each one of $x_1, x_2, \ldots x_n$)
then how many total choice do we have
1st
2nd
3rd
$(n_{C_2})^{th}$
1&2
1&3
1&4
(n-1)&(n)
valid
$2^{\wedge}$ choices w.r.t symmetric Rel
{(a,b)(b,a)}
{(a,b)(b,a)}
2 * 2 * 2 * - - - 2 = $2^{n_{C_2}}$

Symmetric Relation = Any number of diagonal order pairs (i.e. $(a=b)$) and Possible Combination w.r.t non-diagonal order pairs (i.e. order pairs of distinct elements)

No. of symmetric Relⁿ = $2^n * 2^{(n_{C_2})}$

$2^{(n_{C_2})}$ ← we can choose two distinct elements out of n elements in n_{C_2} ways

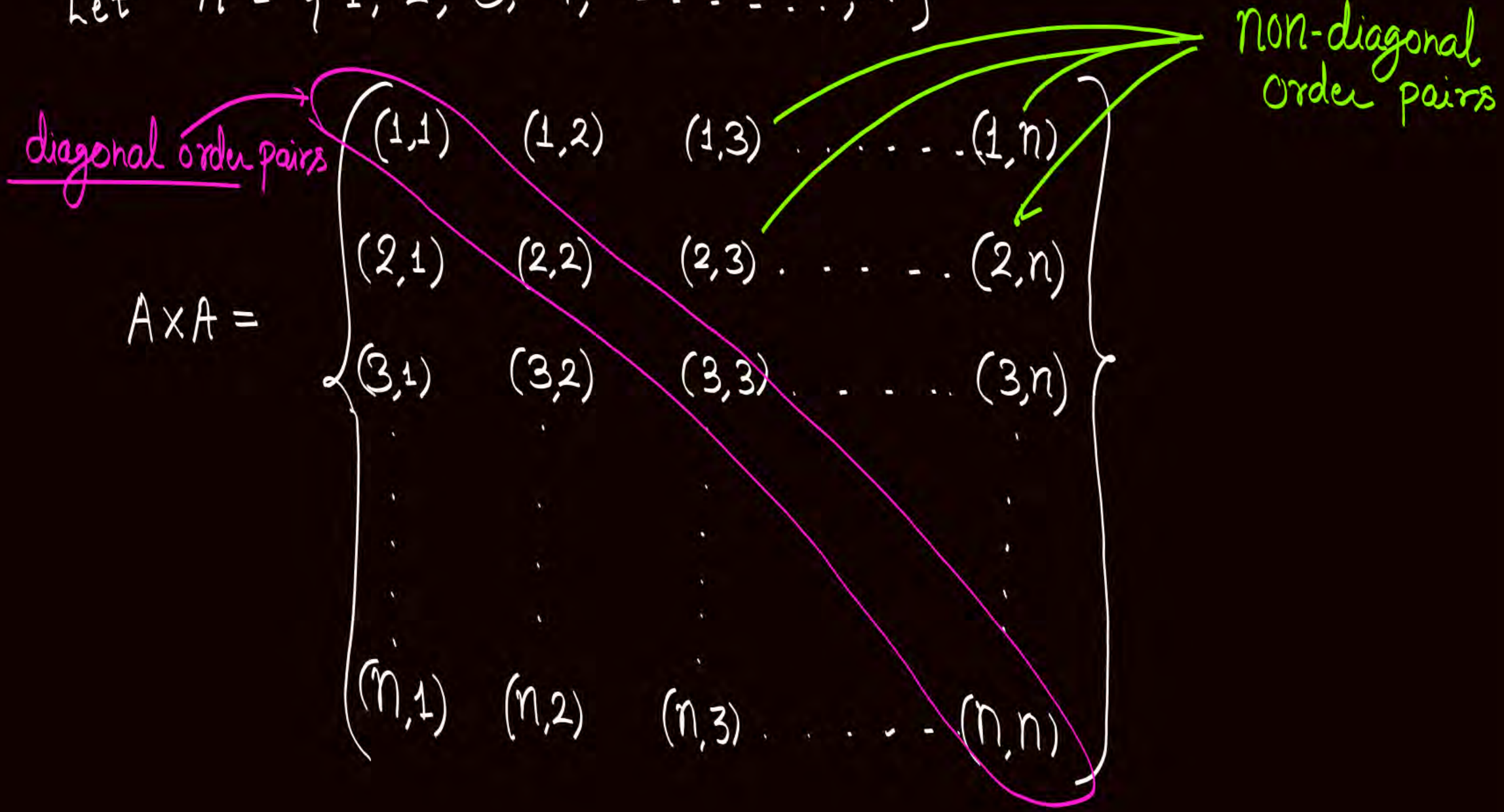
and for each one of this pair of distict elements we have two choice of order pairs w.r.t. symmetric Relⁿ

$= 2^n * 2^{\frac{n(n-1)}{2}}$

$= 2^n * 2^{\frac{n^2-n}{2}}$

$= 2^{\frac{n(n+1)}{2}}$

Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$



H.W.

Q: Let $A = \{1, 2, 3, 4, \ldots, n\}$

How many anti-symmetric relation are possible on set A.

Anti-symmetric Relation = Any number of diagonal order pairs and Possible choices wrt non-diagonal order pairs {i.e. wrt order pairs of distinct elements}

Another approach:→

- In a non-diagonal order pair (a, b) we know $a \neq b$ i.e a & b are distinct.
- For two distinct elements 'a' & 'b' possible combination of order pairs of a & b that may be present in a relation are,

(i) both (a,b) & (b,a) are present

(a,b), (b,a)

Not allowed w.r.t anti Symmetric Relⁿ

(ii) (a,b) is present but (b,a) is absent

(a,b) ~~(b,a)~~ (x)

(iii) (a,b) is not present but (b,a) is present

~~(a,b)~~ (x), (b,a)

(iv) Neither (a,b) is Present nor (b,a) is present

~~(a,b)~~ (x), ~~(b,a)~~ (x)

This three choices are allowed w.r.t. anti-symmetric Relⁿ

H.W.

Q: Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

How many anti-symmetric relation are possible on set A.

Anti-symmetric Relation = Any number of diagonal order pairs and Possible choices wrt non-diagonal order pairs {i.e. wrt order pairs of distinct elements}

No. of anti-symmetric Reln $= 2^n * ({}^nC_2) * 3$

$= 2^n * 3^{\frac{n(n-1)}{2}}$

Ans

Two distinct element out of n can be chosen in ${}^nC_2$ ways and for each one of them we have

3 valid Combinations wrt anti-symmetric reln

H.W.

Q: Let $A = \{1, 2, 3, 4, \ldots, n\}$

How many asymmetric relation are possible on set A.

Asymmetric Relⁿ = None of the diagonal order pair should be present **and** Possible Combinations w.r.t non-diagonal order pairs.

Another approach:→

- In a non-diagonal order pair (a, b) we know $a \neq b$ ie a & b are distinct.
- For two distinct elements 'a' & 'b' possible combination of order pairs of a & b that may be present in a relation are,

(i) both (a,b) & (b,a) are present

(a,b), (b,a)

Not allowed in Asymmetric Rel^n

(ii) (a,b) is present but (b,a) is absent

(a,b) $\cancel{(b,a)}$ ✗

(iii) (a,b) is not present but (b,a) is present

$\cancel{(a,b)}$ ✗, (b,a)

(iv) Neither (a,b) is Present nor (b,a) is present

$\cancel{(a,b)}$ ✗, $\cancel{(b,a)}$ ✗

This three cases are allowed w.r.t. Asymmetric Rel^n

H.W.

Q: Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

How many asymmetric relation are possible on set A.

Asymmetric Relⁿ = None of the diagonal order pair should be present **and** Possible Combinations w.r.t non-diagonal order pairs.

No. of Asymmetric Relⁿ $= {}^{n}C_{0} * (3)^{{}^{n}C_{2}}$ } same as anti-symmetric

$= 1 * 3^{\frac{n^2-n}{2}}$

$= 3^{\frac{n^2-n}{2}}$ Ans

Note

1. Every asymmetric relation is anti-symmetric as well, but every anti-symmetric relation need not be asymmetric
2. Every subset of an anti-symmetric relation is also anti-symmetric
3. Every subset of asymmetric relation is an asymmetric relation
4. Empty relation is symmetric, as well as anti-symmetric, as well as asymmetric Relⁿ.
5. Diagonal relation on set A is the largest relation on set A which is symmetric as well as anti symmetric



Topic : Transitive Relation

A relation R on set A is a transitive relation if and only if,

if $(a \, R \, b$ and $b \, R \, c)$ then $a \, R \, c$, $\forall a, b, c \in A$

(b and b: same element)

i.e. if $((a,b) \in R$ and $(b,c) \in R)$ then $(a,c) \in R$, $\forall a, b, c \in A$

Slide

eg:- Let $A = \{1, 2, 3\}$

$R_1 = \{\ \}$ ← it is smallest transitive rel^n on set A

Empty Rel^n

$R_2 = A \times A$ it is the largest transitive relation on set A.

$R_3 = \{(1,1), (2,2)\}$

{ Because of presence of diagonal order pair
there will be no problem w.r.t. transitive relation,
but sometimes absence of some diagonal order pairs
may create problem w.r.t transitivity }

$R_4 = \{(1,2), (1,1)\}$

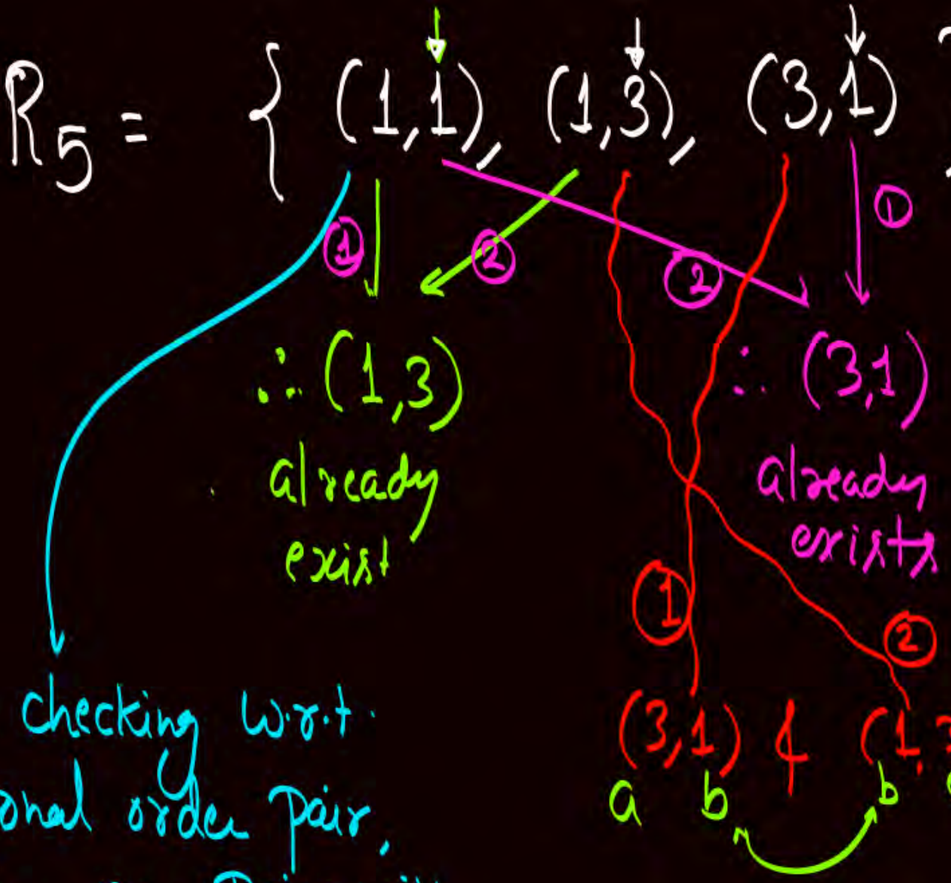
Nothing starts with two

ends with '1'; another pair that starts with '1' is (1,2)

Now (1,1) & (1,2) => (1,2) only which is already present

If we have a diagonal order pair in a relation, and if we check the transitivity property of any other existing pair of the relation with diagonal order pair, then it will never force us to add any new order pair in the rel^n

$R_5 = \{ (1,1), (1,3), (3,1) \} \equiv \{ (1,1), (3,1), (1,3) \}$



While checking w.r.t diagonal order pair, then no new pair will be forced by transitivity

∴ We can skip comparison w.r.t existing diagonal order pairs

$(3,1) \notin (1,3) \Rightarrow (3,3) \notin R_5$

a b b c

$\therefore R_5$ is not a transitive relation

$R_6 = \{(1,2), (1,3)\}$ $\Rightarrow$ Rel^n is transitive Rel^n

ends with '2'
no order pair begin with '2'
∴ if Cond^l false
"don't need"
to worry

ends with '3'
no order pair starts with '3'
∴ if Cond^n is false
& if-then statement is true
∴ No problem

$R_7 = \{(1,2), (2,3)\}$

∴ (1,3)

but $(1,3) \notin R_7$ ∴ Not transitive

Let $A = \{1, 2, 3, \ldots n\}$

We don't generalize the number of transitive relation on set 'A' in terms of 'n'.

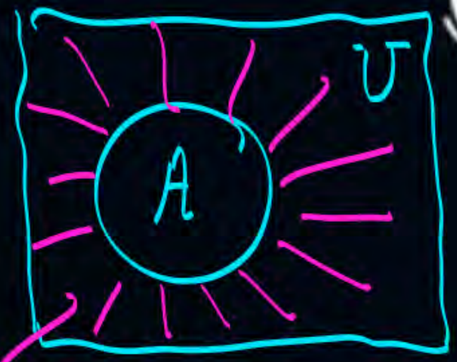


→ Complement is defined w.r.t universal set

Topic : Complement of a relation

Let R be a relation from set A to set B, then complement of relation R is denoted by $\overline{R}$ and it is defined as,

$$\overline{R} = \underbrace{A \times B}_{\text{Universal set w.r.t. relations from A to B}} - R$$



$A^c = U - A$

eg: let $A = \{1,2,3\}$, $B = \{a,b,c\}$

$R = \{(1,a), (1,c), (2,b), (3,c)\}$

$\overline{R} = \{(1,b), (2,a), (2,c), (3,a), (3,b)\}$

Slide



Topic : Inverse of a relation

Let R be a relation from set A to set B, then inverse of relation R {i.e. R^{-1}} will be a relation from set B to set A. and it is defined as

$$R^{-1} = \{(b,a) \mid (a,b) \in R\}$$

from B to A

eg: let $A = \{1,2,3\}$ & $B = \{b,c\}$
& $R = \{(1,b), (2,c), (3,b), (3,c)\}$
is a relation from A to B, then

$R^{-1} = \{(b,1), (c,2), (b,3), (c,3)\}$
is a rel^{n} from B to A.

Slide



H.W.

Topic : Composite of two Relations

Let X,Y and Z be three sets, and

If R is a relation from X to Y, and S is a relation from Y to Z,
then their composition R;S is the relation defined as,

$R;S = \{ (x,z) \mid (x,z) \in X \times Z$ and there exists $y \in Y$ such that $(x,y) \in R$ and $(y,z) \in S\}$

i.e., R;S is a relation from X to Z defined by the rule that $(x,z) \in R;S$ if and only if there is an element
$y \in Y$ such that $(x,y) \in R$ and $(y,z) \in S$



2 mins Summary

Topic Symmetric, anti-symmetric and asymmetric relation

Topic Transitive relation

Topic Complement of a relation

Topic Inverse of a relation

Topic Composite of two relations

THANK - YOU

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